Regional Analysis of Lake County, Illinois

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LAKE COUNTY OVERVIEW

From undeveloped land to a national top-tier, Lake County is a rising county fueled by strategic location and infrastructure development.

History

Used to be an undeveloped prairie, Lake County began transforming in the early 1830s into small villages and farming communities inhabited by the native Potawatomi. The westward expansion from the eastern United States brought settlers in search of new land and better opportunities. By 1834, the area saw its first permanent settlers, and the Caspar Ott Log House was constructed in 1837 (Visit Lake County 2025).

FIGURE 1 THE CASPAR OTT LOG HOUSE

In 1839, Lake County was officially created by the Illinois State Legislature. The name 'Lake' was chosen because of its proximity to Lake Michigan and the presence of approximately forty small lakes within the county's boundary (Brookhaven Press 1877). The county government was established in Waukegan, a name derived from the Potawatomi word for 'Little Fort,' where a courthouse was built due to the community's prospering harbor and lakefront location.

The population surged from about 2,600 residents in 1840 to more than 14,000 by 1850, driven largely by immigrants seeking new opportunities. Throughout the late nineteenth century, the population continued to grow, and the county gained national recognition as a place of promise, especially from World War II. In 2025, Lake County is home to 723,001 residents, making it the third most populous county in Illinois due to its proximity to downtown Chicago in Cook County. Waukegan remains the largest city in the county with 89,900 residents, while Old Mill Creek is the smallest, with a population of 159 (World Population Review 2025).

Geography

Lake County is a region in northeastern Illinois with a population density of 1,628 people per square mile. It is bordered by the state of Wisconsin to the north, Lake Michigan to the east, Cook County to the south, and McHenry County to the west. The county spans 460 square miles (294,400 acres) with a breadth of 19.5 miles. Of its total area—1,368 square miles—444 square miles consist of land, and 935 square miles are water. This means that approximately 67 percent of the county's area is water, largely due to its Lake Michigan frontage.

Lake County also contains significant forested land that provides regional benefits such as groundwater filtration and flood control. Residents and visitors can enjoy various

recreational amenities, including extensive local parks. The Fox River Waterway is one of the county's most important ecological assets and is noted as the busiest inland recreational waterway per acre in the United States (Lake County Partners 2025a).

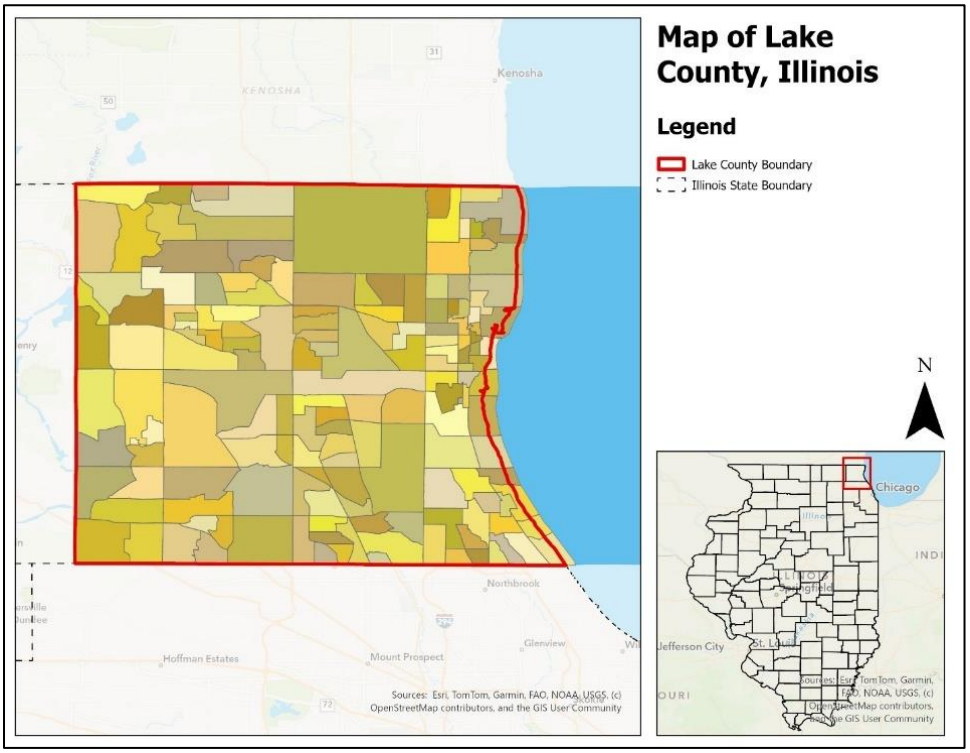

FIGURE 2 MAP OF LAKE COUNTY, ILLINOIS

Transportation Development

Transportation has contributed to Lake County's development. The railroad that commenced in 1855 connected Chicago and Milwaukee via Waukegan served as an important link in the regional transportation network. In 1891, the establishment of Outer Belt Line (the Elgin, Joliet, and Eastern Railroad) provided an intraregional freight carrier that triggered industrial growth (Ebner 2025). The North Shore Line (the Chicago, Milwaukee and North Shore Electric Railway) transported both passengers and freight from 1899 to 1963, further supporting industrial expansion and contributing to the establishment of two major federal defense facilities, Fort Sheridan in Highwood (1887) and the Great Lakes Naval Training Station in North Chicago (1911).

Roadway establishment also encouraged motor vehicles in 1920s. Sheridan Road, a route along Lake Michigan, served as automobile traffic. In 1931, the Skokie Valley Highway (U.S. 41) was commenced and reinforced westward development. This construction contributed to the county's population number to 100,000 residents in 1930 with early urban clusters in Libertyville and Barrington. Besides that, the construction of north-south segment of Tri-State Tollway (I-94) was completed in 1958. This road provided access to O'Hare Airport, major employment centers in Deerfield, Gurnee, and Vernon Hills, and prominent destinations such as Great America amusement park (1976), Hawthorne Center (1973), and Gurnee Mills (1991).

Today, Lake County has various modes of public transportation. Ride Lake County provides paratransit service for the elderly and people with disabilities. Metra operates commuter rail service throughout northeastern Illinois, while Pace Suburban Bus (Pace) offers affordable, innovative, and environmentally responsible bus services across the region. The Regional Transportation Authority (RTA) oversees financing, secures funding, and conducts transit planning for these systems. (Lake County Division of Transportation 2025). The service coverage of Metra and Pace is illustrated in Figures 3 and 4 below.

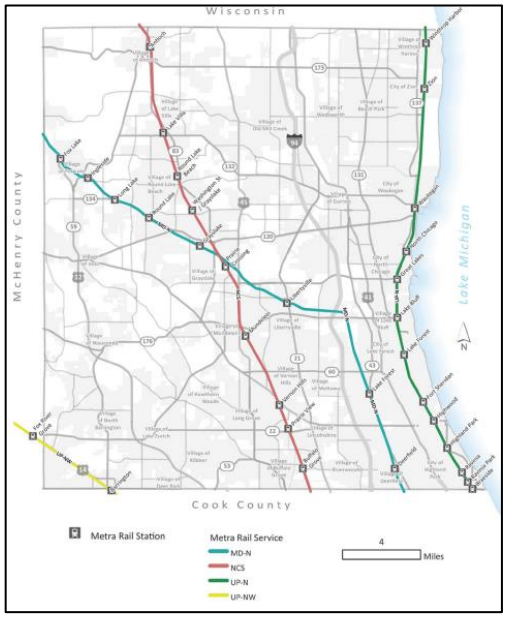

FIGURE 3 METRA RAIL SERVICE MAP
SOURCE: LAKE COUNTY DIVISION OF TRANSPORTATION

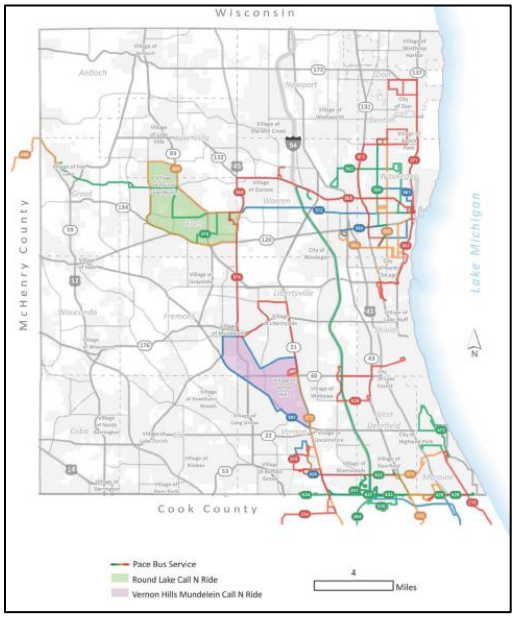

FIGURE 4 PACE SERVICE MAP
SOURCE: LAKE COUNTY DIVISION OF TRANSPORTATION

Economic Development

With the average per capita income of $61,564 and a median household income of $108,917, Lake County is a center of business growth and regional innovation. It is home to a diversified range of industries, from corporate management, transportation, construction, to high-tech industries, which makes this county an attractive destination for job seekers. (Lake County Partners 2022). There are leading employers such as Abbott, Walgreens, AZ Polymers, Sysmex Corporation, and Wintrust Financial Corporation that contribute to strong workplace cultures and support forward-thinking development strategies. This condition attracts employers to grow their business there. During that year, Lake County created 541 new jobs, retained 667 jobs, and received $251 million in capital improvement investment (Sadin 2025). This makes Lake County positioned as the center of Chicagoland's economic growth. Moreover, in 2024, *Site Selection Magazine* ranked Lake County 10th among the top counties in the United States based on the county's performance in attracting corporate facility projects, capital investment, and job creation (Lake County Partners 2025b).

Climate

Lake County has moderate humidity levels throughout the year. Summers are typically hot and humid, with temperatures reaching the mid-90s, while winters are cold and snowy. The county reaches an average of 36 inches of rainfall annually, 39 inches of snow per year, and averages 189 sunny days per year (Best Places 2025). June, August, and September are considered the most pleasant months in Lake County, whereas January and December are the least comfortable. According to the BestPlaces Comfort Index, the county scores 6.8 out of 10, suggesting that overall comfort levels are slightly lower than ideal. An illustration in the figure below shows that, when compared to the broader region, Lake County's climate does not differ significantly.

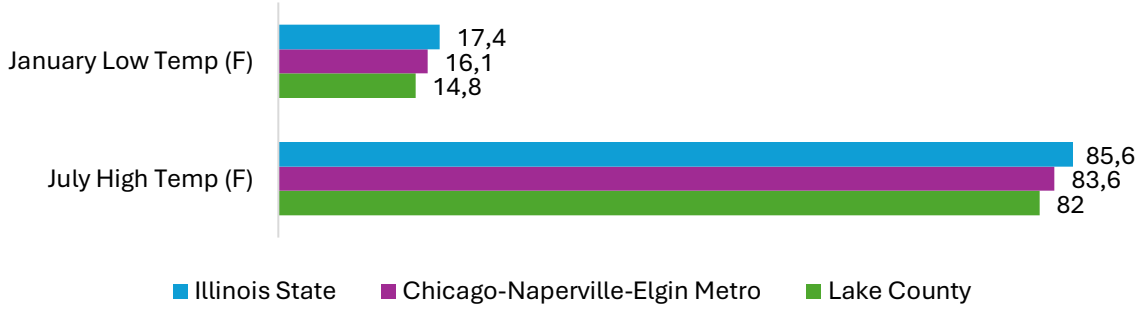

FIGURE 5 CLIMATE TEMPERATURE

Challenges

Despite its strong economic performance, Lake County faces several challenges related to workforce expansion and the sustainability of its commercial assets. The county is struggling to attract younger demographics as baby boomers begin to retire, leaving some communities under-resourced in essential services (Lake County Partners 2025a). To maintain long-term economic growth, Lake County must address issues related to job availability to support workforce renewal, as well as the need for additional housing to accommodate future generations of workers. The county also faces challenges in office space utilization. Demand for traditional office environments—particularly older, single-use buildings—has softened due to the rise of hybrid work models. In contrast, mixed-use districts have demonstrated greater resilience, as they offer workers convenient access to housing, dining, and recreation within walkable environments, creating a more flexible and attractive setting (Lake County Partners 2025a).

From an environmental standpoint, Lake County's proximity to Lake Michigan and its extensive network of inland water bodies make the region particularly vulnerable to flooding around watershed areas. This poses risks not only to housing and commercial development but also to critical road and rail infrastructure essential for regional mobility and goods movement (Lake County Partners 2025a). Winter snow events further threaten infrastructure, including power and telecommunications lines, compounding the county's environmental challenges.

METHODS AND DATA SOURCES

The regional analysis of Lake County, Illinois, uses quantitative methods to describe and examine population, demographic, and economic patterns in the county. The analytical techniques include descriptive analysis, spatial analysis, regression modelling, and forecasting. RStudio and Microsoft Excel are used for data processing, visualization, and map-making. The data are derived from the American Community Survey (ACS) 5-year estimates, OpenStreetMap, United States Census Bureau, CDC Wonder, and U.S. Department of Commerce. The analysis is conducted at multiple scales, ranging from the census tract level to the county level.

POPULATION ANALYSIS

Lake County is projected to have a working-age population concentrated in the western and southern areas.

Population Growth and Projection

The population of Lake County in 2023 was 708,760. According to the American Community Survey (ACS), the population experienced significant growth from 1969 until now. From 1969 to 2023, the population increased by 89.2 percent from 374,447 people to 708,760 people. As shown in Figure 6, the most significant growth occurred between 1985 and 2005 as there was an increase of 46.9 percent from 465,821 to 684,419. After 2005, population growth began to slow, marking a shift from the rapid suburban expansion of previous decades.

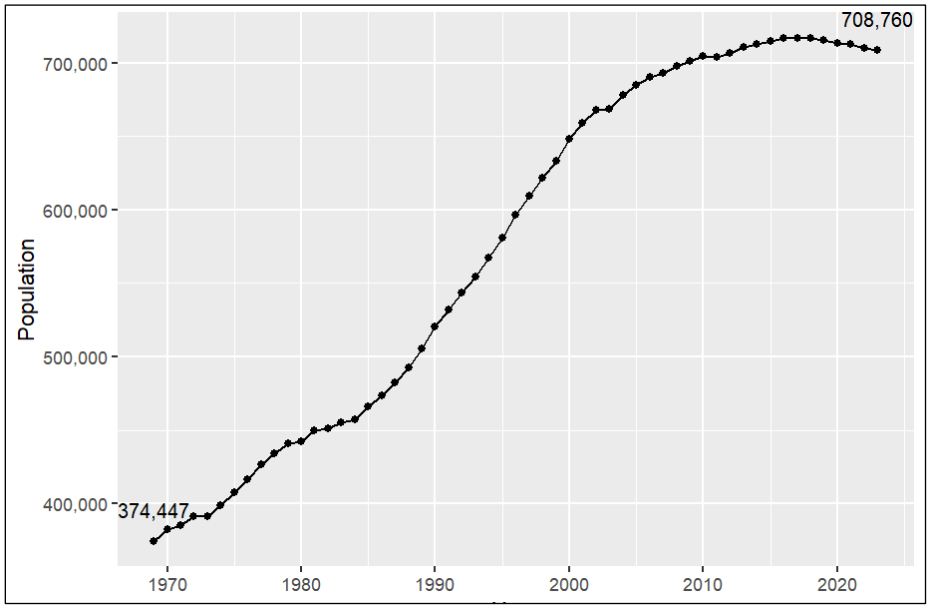

FIGURE 6 POPULATION TREND FROM 1985-2023

The population of Lake County is projected using four methods. The first is a linear model, which estimates population growth through a regression model based on a constant amount. The geometric model projects population growth using a regression model based on a constant growth rate. Next, the constant-change method assumes that the amount of population change observed in the past will continue in the future. Lastly, the percentage-change method assumes that the population growth rate from the previous period will continue.

The result of the project is shown in Figure 7. Among the four methods, the constant change method was selected because its projection appears to be the most reasonable and produces the lowest estimated population. Thus, based on this method, the average annual change is 6,190 people, indicating that Lake County's population is projected to increase by approximately 6,190 residents each year under this model.

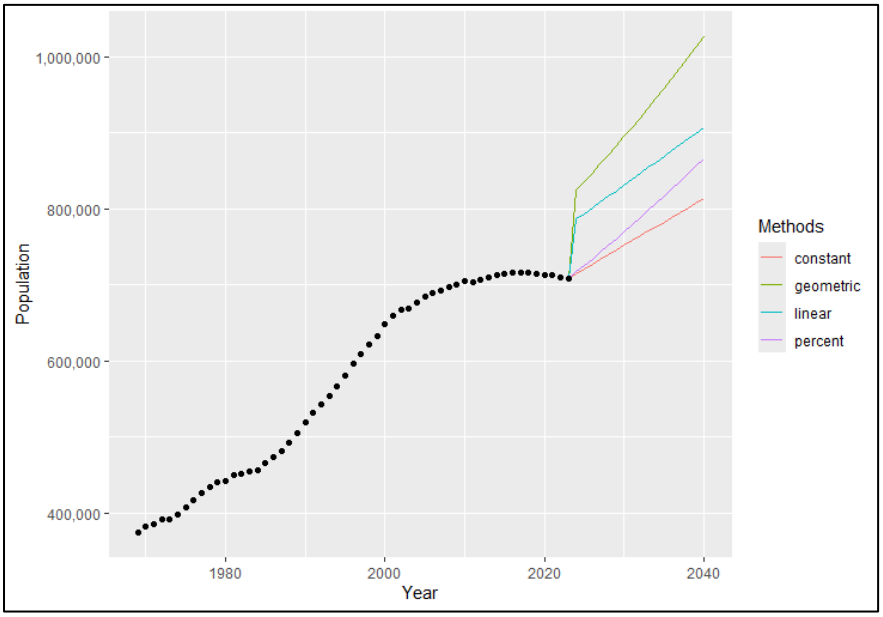

FIGURE 7 POPULATION PROJECTION IN 2040

Population Pyramid

In 2010, Lake County had a total population of 703,462, consisting of 50.1 percent females and 49.9 percent males. As shown in Figure 8, the largest age groups were those aged 45-49 and 10-14, representing 8.3 percent and 8.1 percent of the population, respectively. The smallest age groups were 85 years old and above, comprising only 1.4 percent of the population.

By 2020, Lake County's population increased to 713, 159, with 50.4 percent males and 49.6 females. Figure 9 shows that most age cohorts shifted to the 15-19 and 20-24 age groups, which together accounted for 15.1 percent of the population. Meanwhile, the smallest age groups were aged 80-84 and 85 and above, together making up just 3.5 percent of the total population.

In 2030, Lake County's population is projected to reach 708,375, with 49.3 percent females and 50.6 percent males. Figure 10 shows that the age composition is expected to shift again, with the 15-19 age group projected to represent 6.9 percent of the population, followed closely by the 20-24 age group at 6.75 percent. The smallest age group will remain individuals aged 85 and above, comprising only 2.3 percent of the total population.

Based on that information, Lake County is in the contracting stage of population growth, which is characterized by a low birth rate, low death rate, and longer life expectancy. From 2010 to 2030, the 15-19 and 20-24 age groups remain the largest cohorts; however, there is a noticeable decline in the subsequent 25-29 and 30-34 age groups, as well as an overall decrease in population. This condition may indicate migration of working-age populations seeking better employment opportunities outside the county. Also, with both a low birth rate and a low death rate, this county has implications for the county's labor force, as fewer working-age residents remain in the area to support both younger and older dependents.

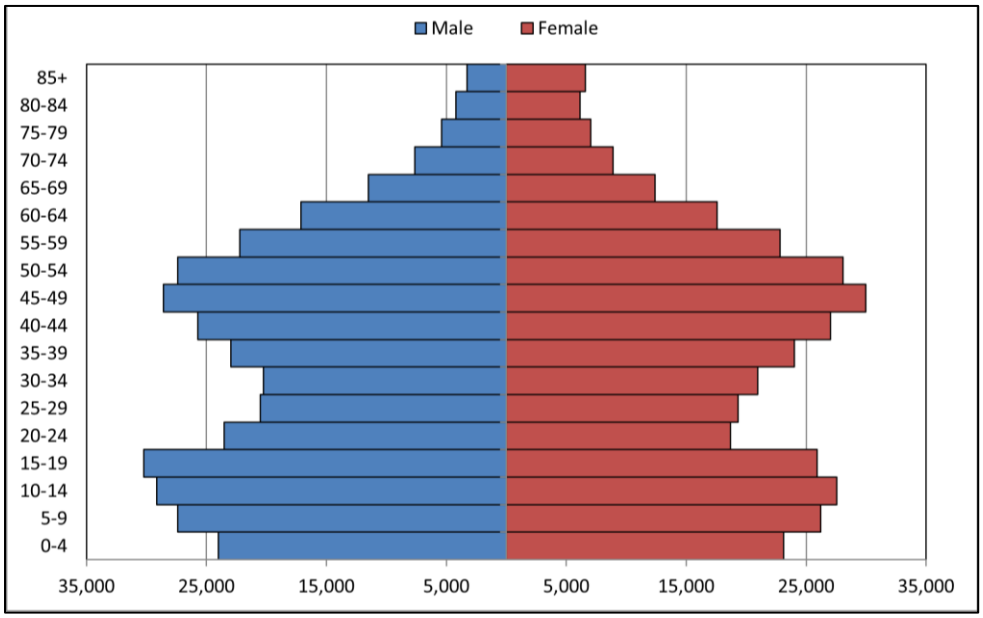

FIGURE 8 POPULATION PYRAMID IN 2010

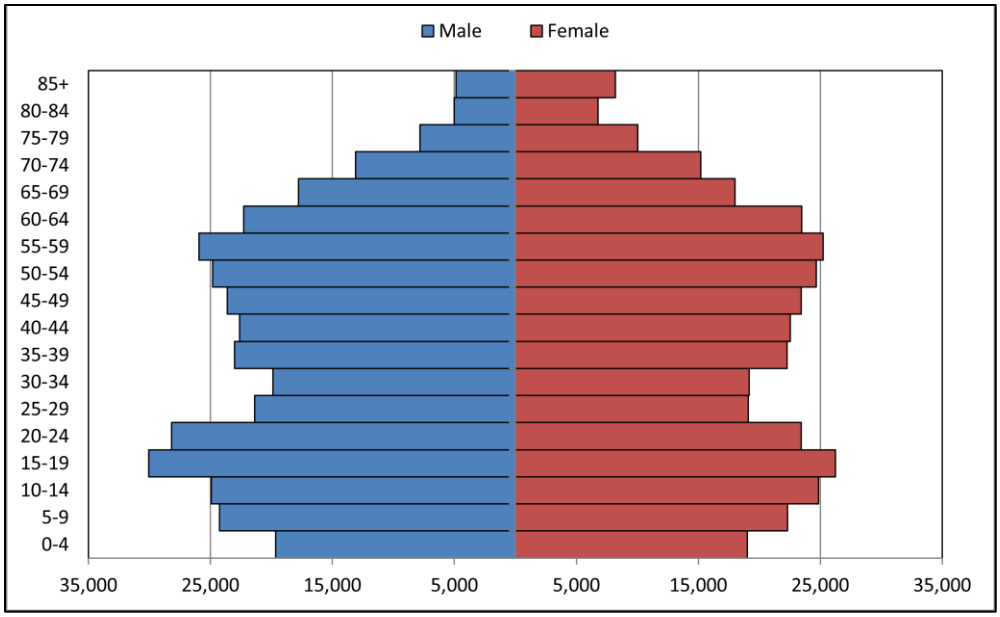

FIGURE 9 POPULATION PYRAMID IN 2020

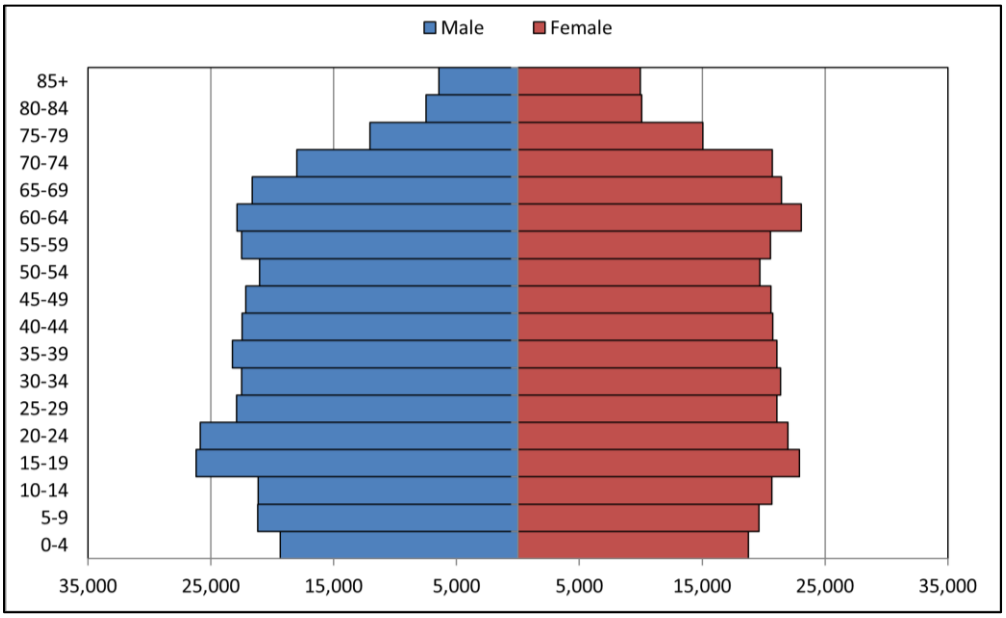

FIGURE 10 POPULATION PYRAMID 2030

Population Density

The population density map in 2020 and 2040 is shown in Figures 11 and 12 below. In 2020, the median population density was 2,220 people per square mile, with densities ranging from 206 – 11,717 people per square mile. By 2040, the median population density is projected to increase to 2,530 people per square mile. The projected minimum density is 235.3 people per square mile and a maximum of 13,352 people per square mile. In 2020, higher-density areas were concentrated in the eastern part of the county, with emerging pockets of density appearing in the western and southern areas. By 2040, these emerging areas become more pronounced, as indicated by the increasingly darker shading.

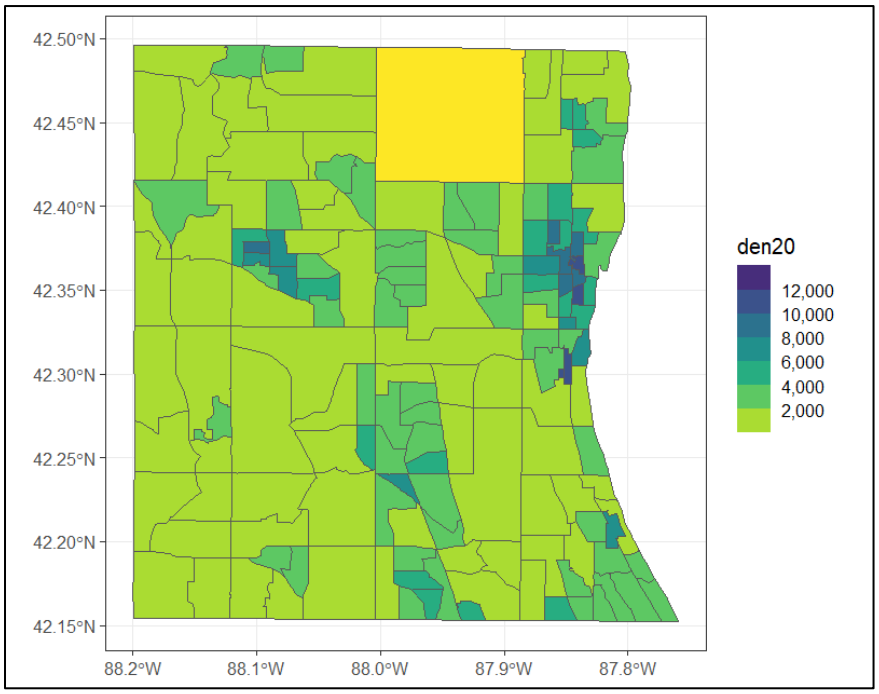

FIGURE 11 MAP OF POPULATION DENSITY IN 2020

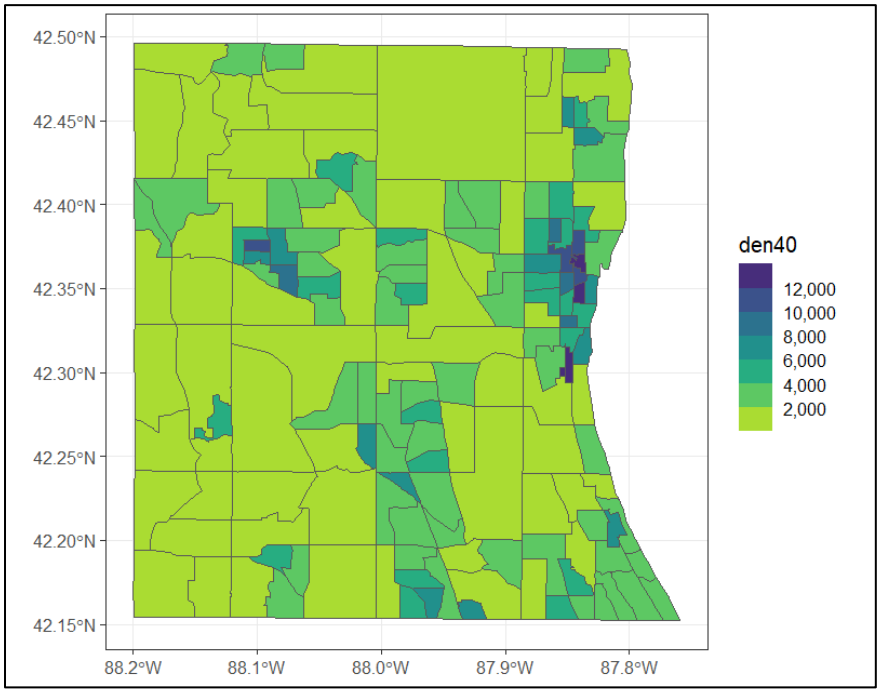

FIGURE 12 MAP OF POPULATION DENSITY IN 2040

DEMOGRAPHIC ANALYSIS

Lake County has a high standard of living and a diverse neighborhood.

Education

The largest share of the population holds a bachelor's degree, accounting for 27.3 percent of residents. In contrast, at the state level, the most common educational attainment category is some colleges, comprising 28.1 percent of Illinois's population. Lake County also has a higher proportion of residents with graduate degrees (19.6 percent) than the statewide rate (14.6 percent). These patterns indicate that Lake County has a more highly educated workforce with advanced skills that align with the needs of firms located in the area. This concentration of human capital is a key factor attracting businesses to invest and operate in Lake County. The comparison of education level between Lake County and the State of Illinois can be seen in Figure 13 below.

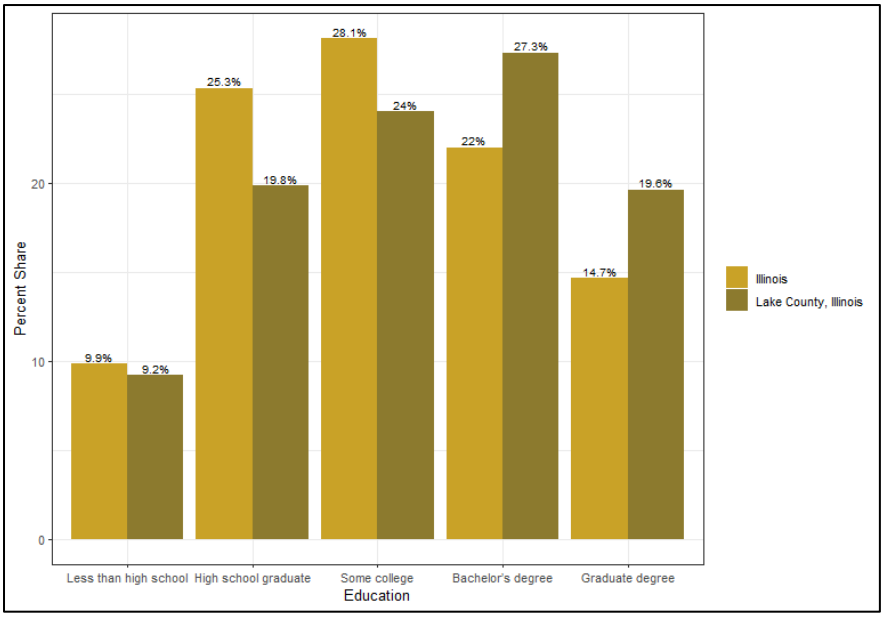

FIGURE 13 EDUCATION LEVEL

Income

Most populations in both the State of Illinois and Lake County have an income of more than $150,000 per year. However, there is a gap between the statewide and county-level proportions. In Lake County, about 33.3 percent of households fall into this income group, compared to 21.5 percent statewide. This disparity suggests that Lake County's higher levels of educational attainment provide access to higher-paying employment opportunities that contribute to greater household earnings relative to the state. A comparison of income distributions between Lake County and the State of Illinois is presented in Figure 14.

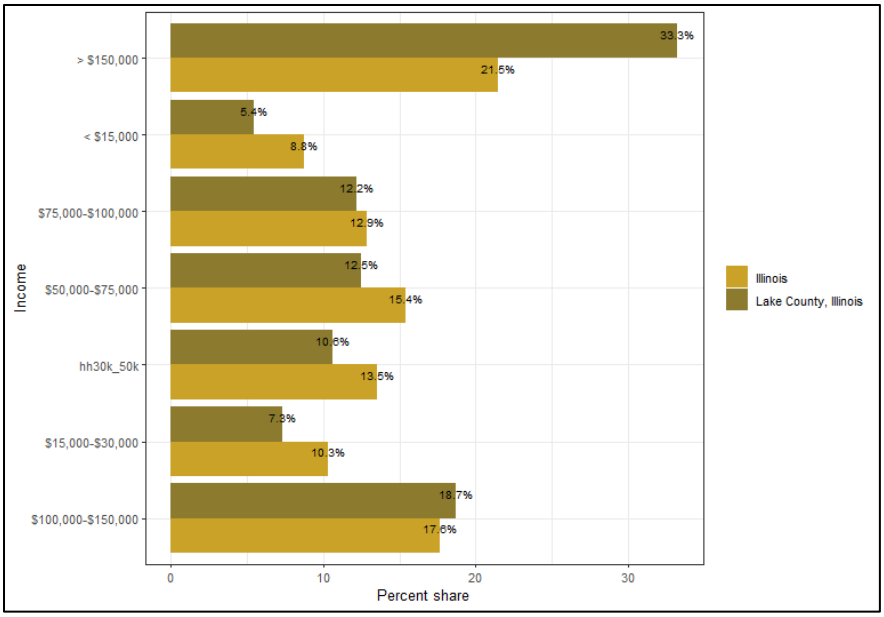

FIGURE 14 INCOME

Poverty

The median poverty rate in Lake County is 5.67 percent, with values ranging from a low of 0.29 percent to a high of 39.27 percent. Compared to the state average, Lake County has a smaller share of residents living below the poverty line at 8.1 percent, whereas Illinois reports that 11.7 percent of its population below the poverty line. This condition indicates that Lake County has high living standards supported by a high level of education and high income of the residents. Figure 15 presents the spatial distribution of poverty in Lake County. As shown on the map, the poverty rate appears to be relatively evenly distributed. Nevertheless, higher poverty rates tend to cluster in the northeastern part of the county, while lower poverty rates are more commonly found in the southern areas.

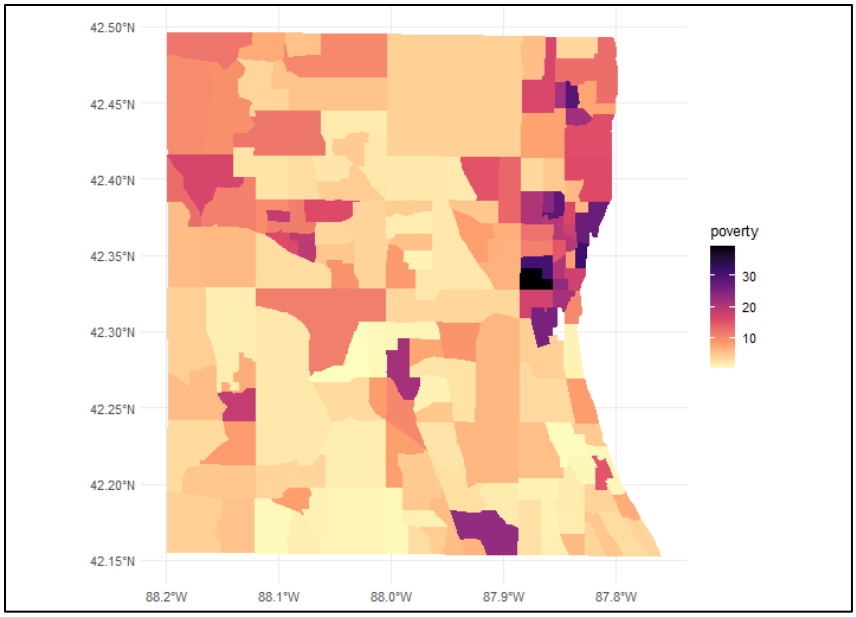

FIGURE 15 MAP OF POVERTY RATE IN 2030

Race/Ethnicity Composition

Lake County comprises at least five racial groups: Hispanic (162,456 people), non-Hispanic White (420,382 people), Black (46,314 people), Asian (59,847 people), and other races (24,160 people). Figure 16 shows that among the total population, non-Hispanic Whites constitute the majority, accounting for 58.9 percent, followed by Hispanics at 22.8 percent. This pattern is similar to that of the State of Illinois, where, among 12,757,634 residents, the majority are non-Hispanic Whites (59.6 percent), followed by Hispanics (17.8 percent). However, compared to the state, Lake County has a higher proportion of Asian and Hispanic ethnic groups, indicating a greater racial diversity.

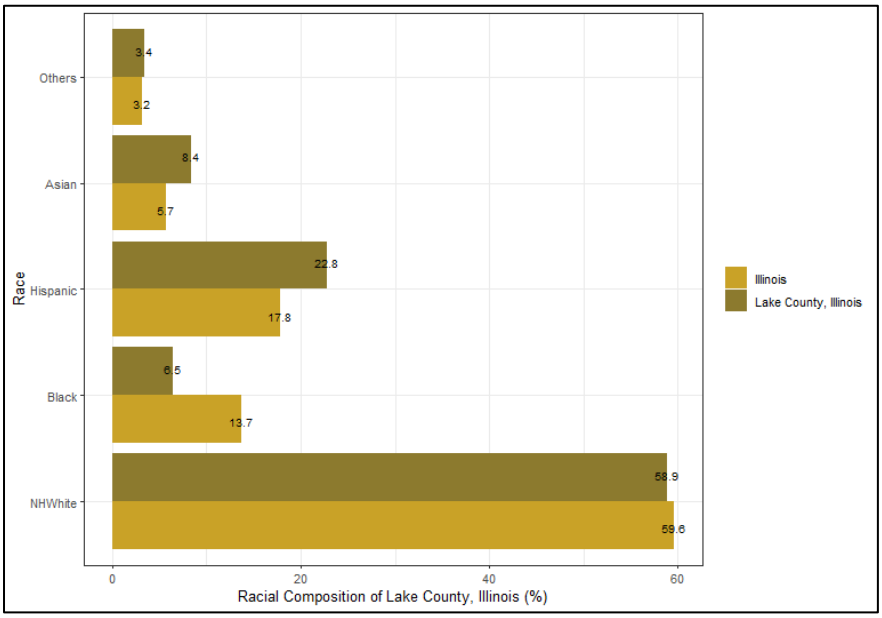

FIGURE 16 RACIAL COMPOSITION

Figure 17 shows the population distribution by race in Lake County. Each point on the map represents 100 people. The distribution pattern shows that non-Hispanic Whites are almost spread across the entire county but are more concentrated in the southern part. Meanwhile, other races are clustered in specific areas. For instance, Hispanics (in green dots) are primarily concentrated in the western part of Lake County, with smaller clusters in the eastern part.

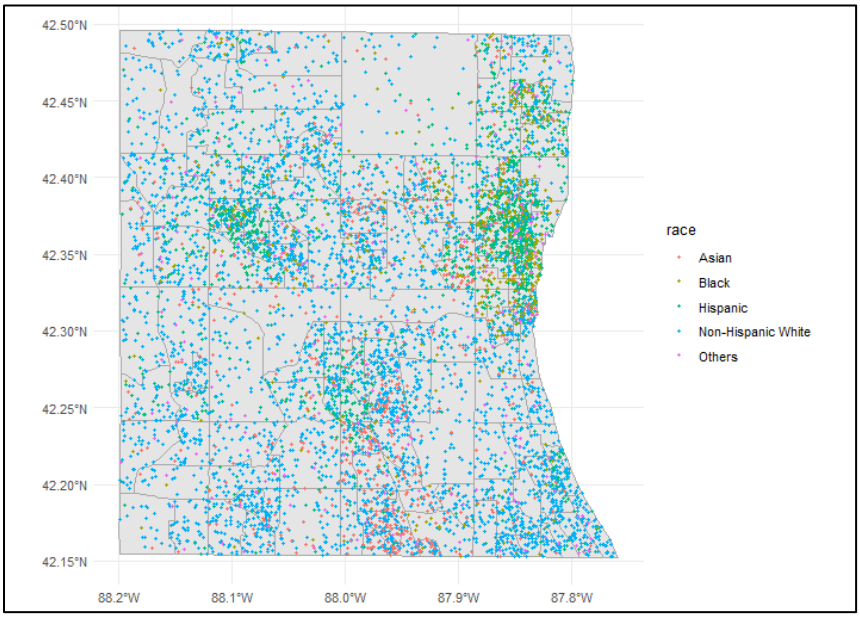

FIGURE 17 RACE DISTRIBUTION MAP

Household Income and Home Value

Figure 18 shows the red line is the median home value trend, and the green line is the median household income trend. In Lake County, the median home value has consistently been higher than the median household income over time. Moreover, home values have increased substantially, while income growth has not kept pace.

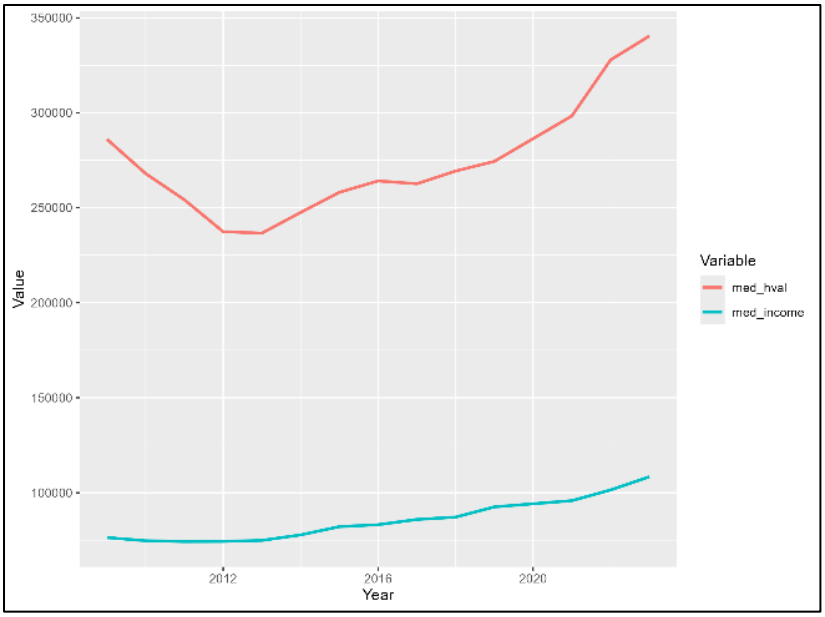

FIGURE 18 MEDIAN HOUSEHOLD INCOME AND HOME VALUE FROM 2009-2023

Figures 19 and 20 below visualize the median household income and home value at the census tract level. As shown in Figure 19, median household incomes are generally higher in the southern part of Lake County and lower in the northern part, with values ranging from $29,113 to $250,001. A similar spatial pattern appears in median home values, shown in Figure 20. Home prices tend to be highest in the southern portion of the county and become more affordable moving northward. Median home values range from $108,200 to $1,113,400. This indicates that higher-income households who can afford relatively high-value homes are concentrated in this area.

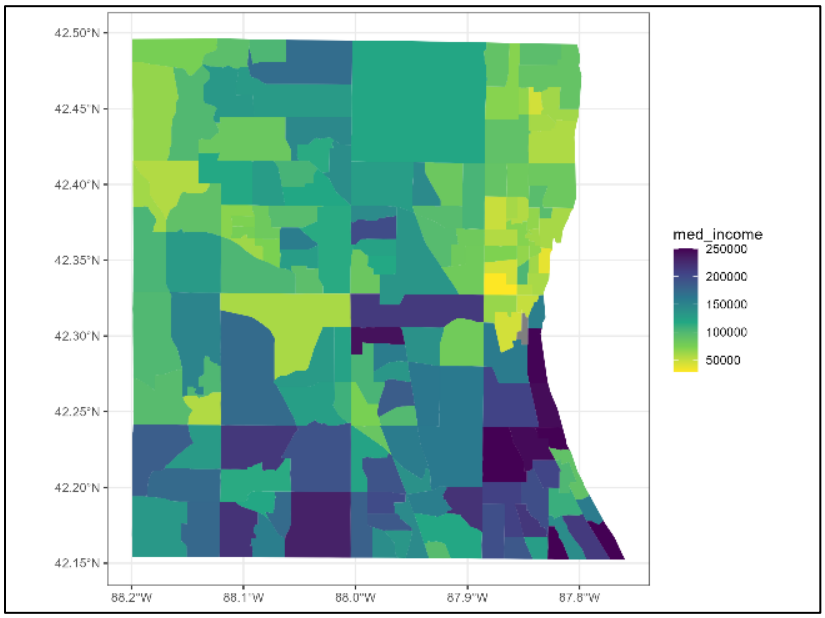

FIGURE 19 MAP OF MEDIAN HOUSEHOLD INCOME

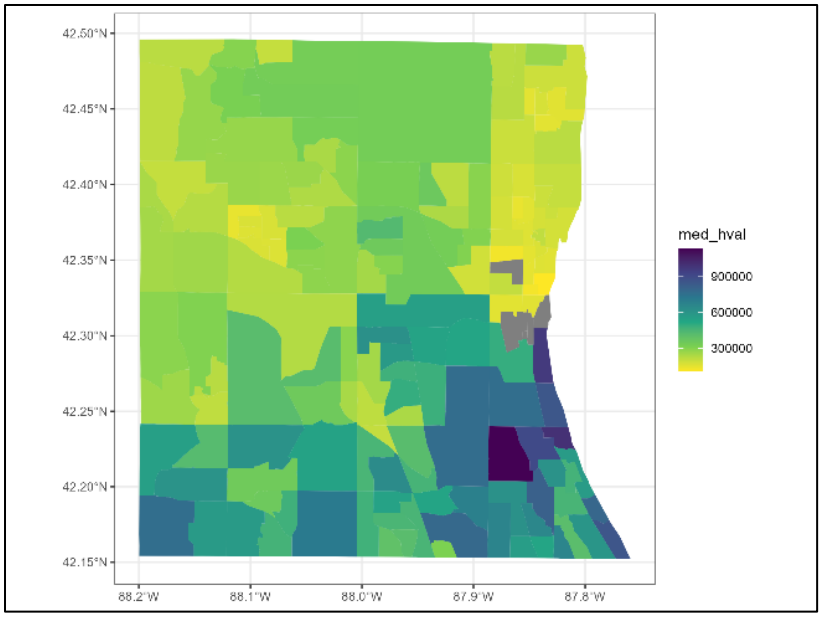

FIGURE 20 MAP OF MEDIAN HOME VALUE

ECONOMIC ANALYSIS

Economic Structure and Growth

The economic structure and growth of Lake County are described using the location quotient (LQ) and the shift-share analysis. According to LQ analysis, industry specialization is considered if several industries have an LQ more than 1. From table 1 below, the top five industries include military (3.90), management of companies and enterprises (1.88), manufacturing (1.80), wholesale trade (1.62), and finance and insurance (1.33). LQ with values more than 1.2 represents the most highly specialized industries. These sectors serve as Lake County's basic industries as they produce goods and services beyond local demand and contribute to economic exports.

Lake County functions as a job generator both within the region and relative to national trends

Lake County had an employment multiplier of 6.68, meaning that for every one new job created, approximately six additional jobs are generated. Manufacturing is an industry that generates income for the county's economy as it produces goods and services for export markets and contributes 24,596 jobs. However, the sector that supports local employment is retail trade, as it provides 44,504 jobs for the local area.

TABLE 1 LQ ANALYSIS

Industry	LQ	Base (Export) Employment	Non-base (Local) Employment
Military	3.90	12,349	4,263

Industry	LQ	Base (Export) Employment	Non-base (Local) Employment
Management of Companies and Enterprises	1.88	5,941	6,738
Manufacturing	1.80	24,596	30,848
Wholesale Trade	1.62	9,486	15,414
Finance and Insurance	1.33	9,776	29,613
Utilities	1.23	320	1,381
Administrative and Support and Waste Management and Remediation Services	1.19	5,697	29,786
Retail Trade	1.08	3,646	44,504
Professional, scientific, and technical services	1.02	754	36,447

The result of the shift-share analysis is presented in Table 2 below. There are three industries that have the highest total contributions: transportation and warehousing, finance and insurance, and real estate and leasing. Overall, the transportation and warehousing sector contributed 11,302 jobs to Lake County's employment. The sector is also the most competitive, as it adds 4,963 additional jobs beyond what would be expected from national trends alone, but it would have added 1,242 jobs if it had grown at the same rate as the national rate. Through industry mix effects, this sector would generate 5,097 jobs, indicating strong performance.

In contrast, the construction sector is the largest contributor to job loss, with a total decline of 5,303 jobs. This sector shows weak competitiveness as it loses an estimated 6,317 additional jobs relative to expected trends. Although this sector would have added 4,527 jobs if it had grown at the same rate as the national scale, it shows a decline of 3,513 jobs from an industry mix effect.

TABLE 2 SHIFT-SHARE ANALYSIS

Industry	Growth Effect (NG)	Mix Effect (IM)	Regional Shift (RS)	Sum
Farm + Forestry and fishing	194	-187	-161	-154
Mining, quarrying, and oil and gas extraction	152	-121	-549	-518
Utilities	92	-70	1178	1,200
Construction	4,527	-3,513	-6,317	-5,303
Manufacturing	9,666	-13,085	6,110	2,691
Wholesale trade	5,042	-4,273	-3,384	-2,615
Retail trade	9,573	-8,080	-5,586	-4,093
Transportation and warehousing	1,242	5,097	4,963	11,302
Information	1,007	-534	-1,862	-1,389
Finance and insurance	5,615	9,003	-5,874	8,744
Real estate, rental, and leasing	3,533	6,614	-4,680	5,467
Professional, scientific, and technical services	6,402	5,493	-9,631	2,264
Management of companies and enterprises	1,620	2,858	-638	3,840
Administrative and support, waste management, and remediation services	5,933	-84	-2,746	3,103

Industry	Growth Effect (NG)	Mix Effect (IM)	Regional Shift (RS)	Sum
Educational services	1,657	1,139	-2,172	624
Health care and social assistance	6,369	4,329	-9,438	1,260
Arts, entertainment, and recreation	2,033	223	-3,289	-1,033
Accommodation and food services	4,417	534	-3,153	1,798
Other services (except government and government enterprises)	4,152	-910	-3,264	-22
Federal civilian	1,001	-728	-496	-223
Military	3,268	-4,778	290	-1,221
State and local	6,727	-6,723	-771	-767
Total	84,220	-7,796	-51,469	24,955

Employment Center

Determining the employment center in Lake County requires population and employment density data, which in this analysis uses 2022 data. The density maps are shown in Figure 21 and Figure 22 below. The median population density was 4,256 people per square mile, with values ranging from 1,050 to 11,000 people per square mile. High-density areas were concentrated in the western and eastern portions of the county, with emerging clusters in the southern part of Lake County. The median employment density was 731 jobs per square mile, with densities ranging from 21 to 17,446 jobs per square mile. High-employment-density areas were primarily located in the southern part of the county, with emerging clusters extending toward the northern portion of Lake County.

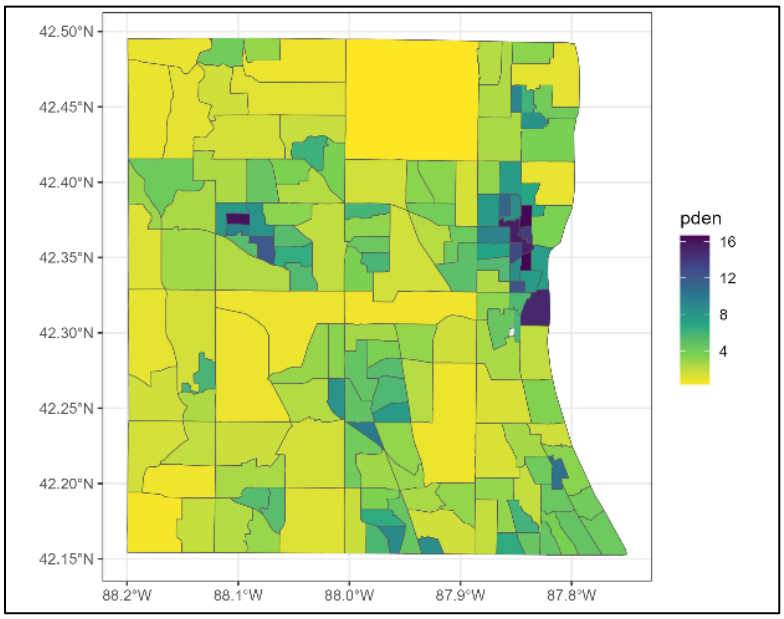

FIGURE 21 POPULATION DENSITY IN 2022

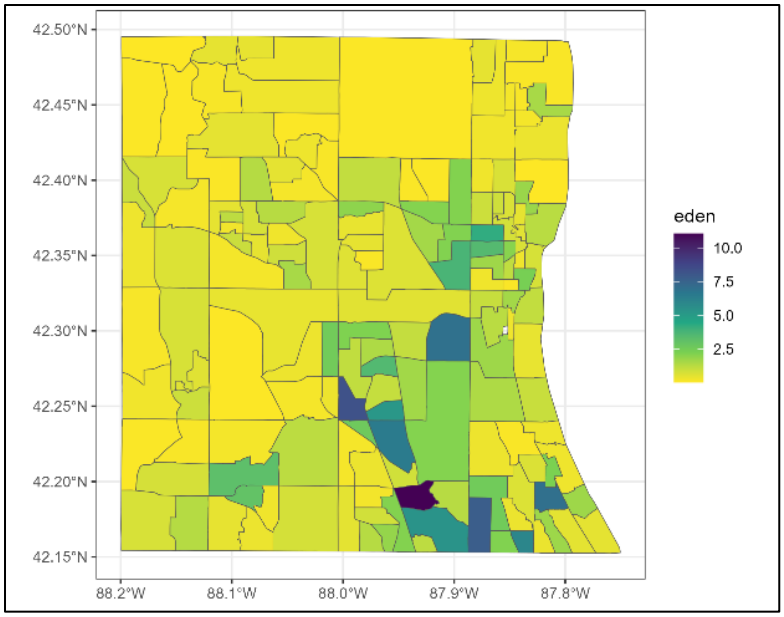

FIGURE 22 EMPLOYMENT DENSITY IN 2022

The employment center candidates in Lake County were identified using percentile rankings of employment density. Tracts at or above the 90th percentile were selected, with the 90th-percentile employment density serving as the minimum threshold required for an area to qualify as an employment center. The resulting employment center candidates are shown in Figure 23. Furthermore, Figure 24 shows eight center candidates that were identified using queen contiguity, a method in which two polygons are considered contiguous if they share even a single point. The CBD is purple in color due to the highest share of employment, while the remaining candidates were classified as subcenters according to their relative employment proportions. The classification is summarized in Table 1 in the appendix.

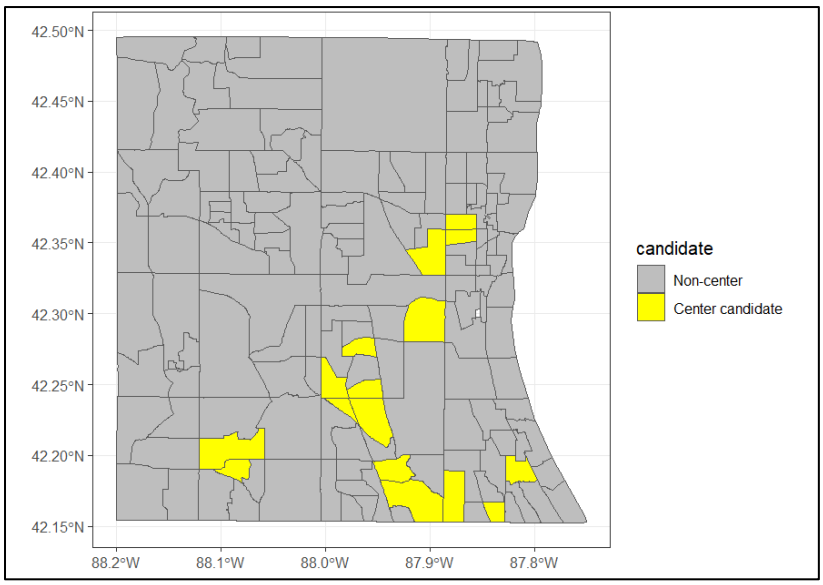

FIGURE 23 EMPLOYMENT CENTER CANDIDATES

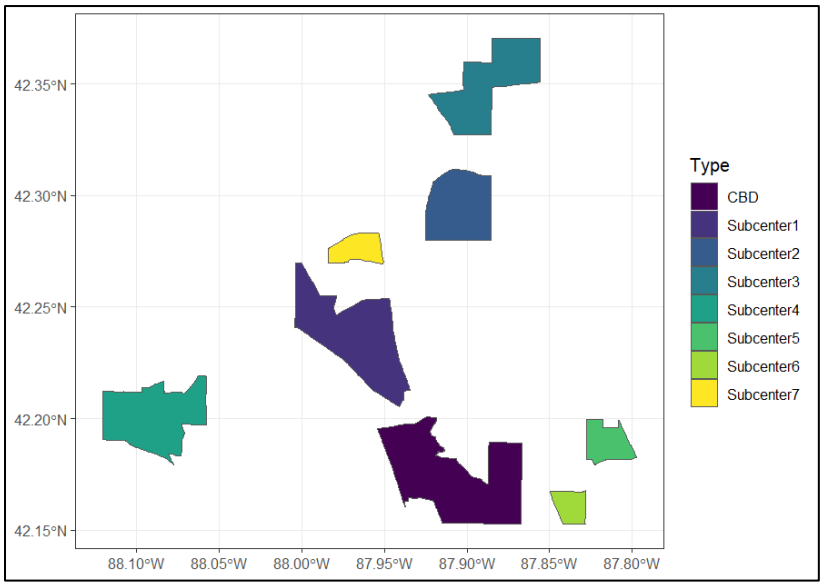

FIGURE 24 DEDICATED EMPLOYMENT CENTER

Density gradient function is used to measure the decrease in density per mile of distance from the employment centers. In this analysis, the employment centers are chosen by the tracts with the highest employment share in each group. Then, regression models are operated to examine how distance from each center influences the spatial distribution of population and employment density. The regression results are presented in the appendix.

For employment density, the model suggests that it decreases by approximately 10.4% for each unit increase in distance from the CBD. This means that the employment activities are concentrated around the CBD, which shows a monocentric urban structure. For population density, the model suggests that it decreases by 8.1% as distance from the subcenters increases, which indicates that areas closer to subcenters tend to have higher population density. However, since the coefficient is not statistically significant, it suggests that the CBD does not play a role in shaping the distribution of population density. The subcenters in Lake County are attracting population, reflecting a decentralized or polycentric pattern.

Housing Affordability

Renters face greater housing affordability challenges than homeowners in Lake County.

The housing affordability in this county fluctuates from 2009-2023, with a notable decline beginning around 2019 and continuing through 2022. Despite this drop, the index remains above 100 in all years, indicating that, on average, the Lake County housing market has remained affordable for residents.

The pattern of the housing affordability index is shown in Figure 25. Census tracts in the northern part of the county generally show higher HAI values than those in the south, indicating that homes are more affordable in the northern areas. In other words, residents in the northern part of Lake County have sufficient income relative to local home prices.

Cost-burden measures are conducted to include both owner- and renter-occupied households and indicate the degree of financial strain associated with housing payments. This measure can also cover the downside of housing affordability index that can only identify owner-occupied homes. The type of cost-burden is divided into two categories: cost-burdened means that occupiers spend 30% of their income to pay for housing cost and severely cost-burdened means that occupiers spend 50% of their income to pay for housing cost. The former group signals financial stress, but the latter group signals critical financial stress because they face severe trade-offs with food and health.

Housing affordability in Lake County poses a serious challenge for renters than for homeowners. The distinction between owner- and renter-occupied units is illustrated in the figure in the appendix. Rent households are more prone to financial strain because the home value is continuously increasing while the income cannot keep up the pace. The group of renters who spend over 30% and 50% of income on housing costs is showing significant increase trend since 2019. These trends indicate that many renters are undergoing financial stress and facing difficult trade-offs in meeting basic needs. In contrast, the group among homeowners who either spend over 30% or 50% of their income on housing costs has generally declined.

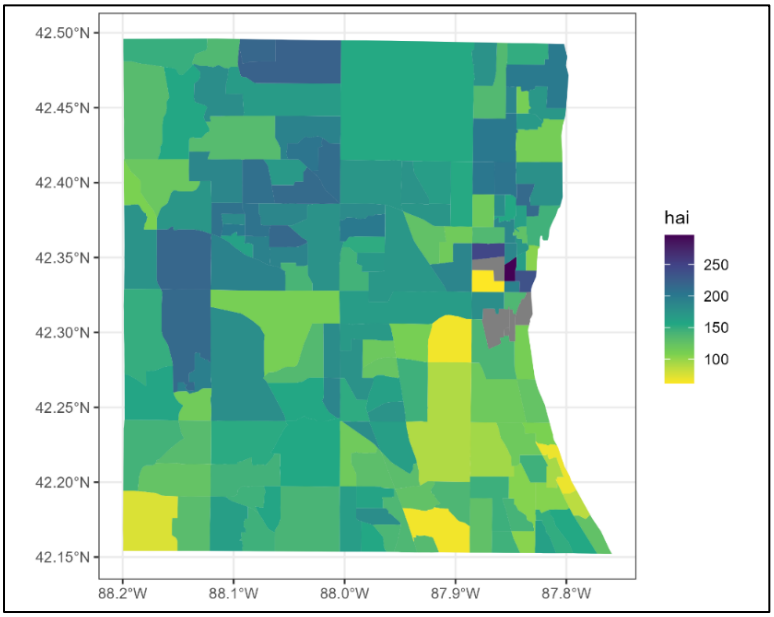

FIGURE 25 MAPPING OF HOUSING AFFORDABILITY INDEX

The distribution among groups at census-tract level can be seen in Figures below. Figure 26 and 28 shows the distribution of percentage of cost-burdened and severely cost-burdened homeowners respectfully, while Figure 27 and 29 shows the distribution of percentage of cost-burdened and severely cost-burdened renters respectfully. Figure 26

shows that only a few census tracts have a significant value share of cost-burdened homeowners. However, Figure 27 reveals that cost-burdened renters are spread across many tracts throughout the county. When residents spend more than 50% of income on housing costs, the maps show similar patterns to the group that spends 30% of income. This evidence strengthens the conclusion that housing affordability challenges exist in Lake County, with renters experiencing the greatest impact.

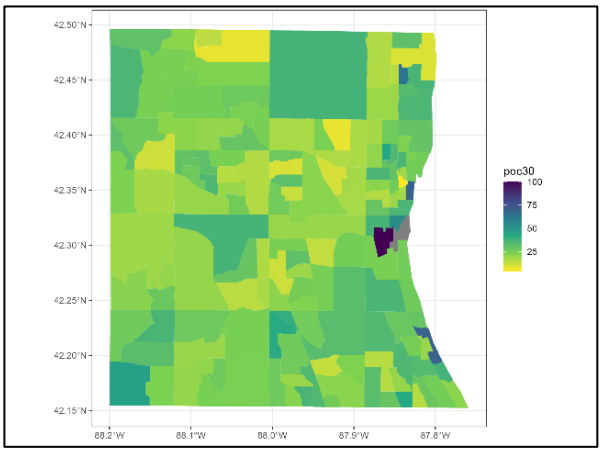

FIGURE 26 PERCENTAGE OF COST-BURDENED HOMEOWNERS

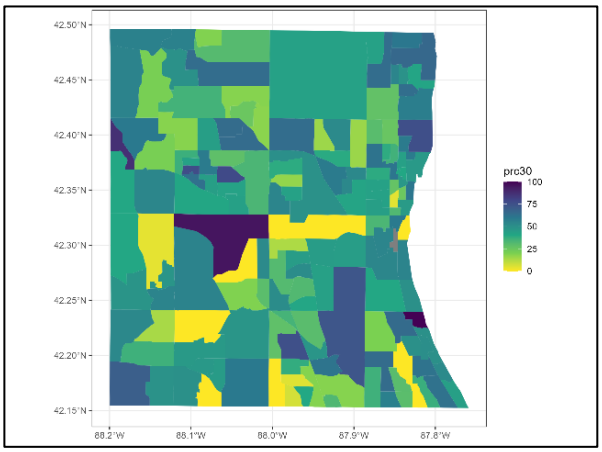

FIGURE 27 PERCENTAGE OF COST-BURDENED RENTERS

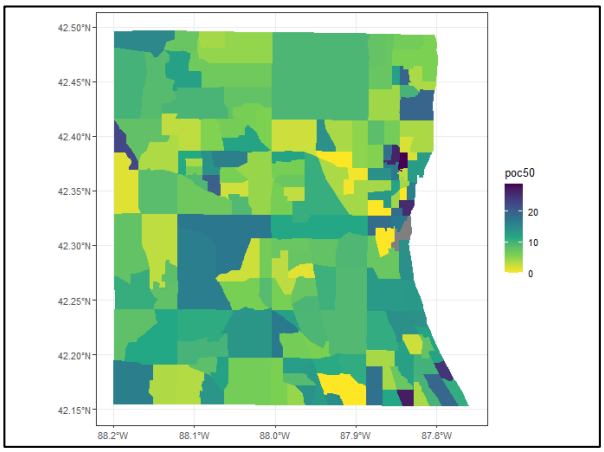

FIGURE 28 PERCENTAGE OF SEVERELY COST-BURDENED HOMEOWNERS

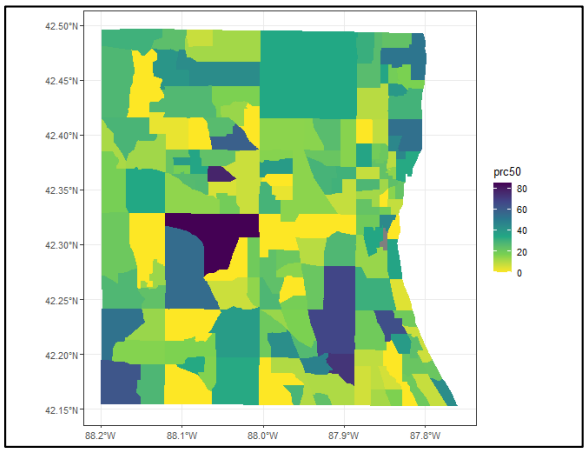

FIGURE 29 PERCENTAGE OF SEVERLY COST-BURDENED RENTERS

Transportation Accessibility

Accessibility is highest in areas near the central business district.

The transportation accessibility analysis uses two approaches: cumulative opportunity measure of accessibility and gravity-type measure of accessibility. The former

approach counts the number of opportunities reached within a certain travel time, which in this analysis the time is within 30 minutes. To substitute the shortage of this approach that the way opportunities are perceived and deal with cut-off time problem, the latter approach is used. It measures the spatial interaction potential between an origin and multiple destination. This methodology weighs the accessible opportunities and integrates a distance-decay function into its formulation (Kapatsila et al. 2023).

The measurement from cumulative approach in Figure 30 shows that dark purple areas have high accessibility, indicating that residents can reach many jobs within a given distance. Whereas yellow areas have low accessibility, meaning residents can access fewer jobs within the same distance threshold. The central-wester part of Lake County exhibits the highest job accessibility, while accessibility declines as distance from job centers increases.

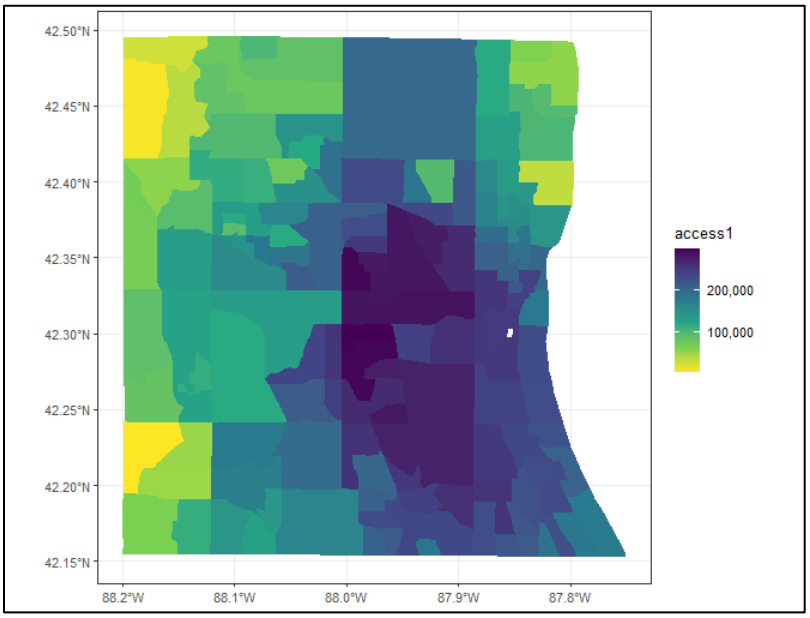

FIGURE 30 CUMULATIVE OPPORTUNITY MEASUREMENT

Figure 31 shows a different pattern of accessibility. High accessibility is concentrated in the central-southern part of Lake County, which is identified as the CBD, indicating a strong gravitational pull of jobs in this area. Accessibility is also observed in several subcenters extending toward the northern part of the county. As distance from both CBD and subcenters increases, accessibility declines, as indicated by the shifting color to yellow areas representing lower accessibility.

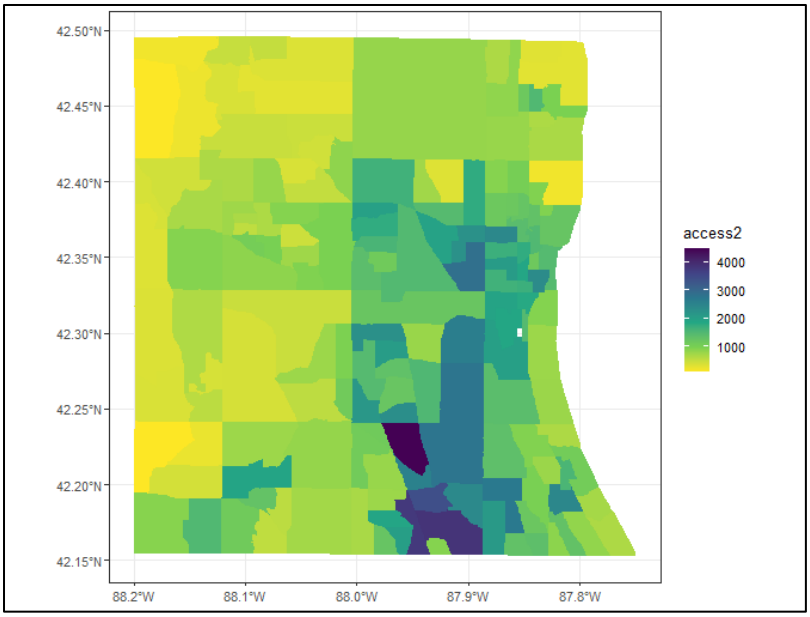

FIGURE 31 GRAVITY-TYPE MEASUREMENT

CONCLUSION

The regional analysis of Lake County, Illinois, covers various aspects, including population, demographics, and the economy. Lake County used to be an undeveloped prairie and small village, but due to its strategic location, specifically near Cook County where Chicago City lies, and transportation development, it is now a center of business growth and regional innovation that is attractive for job seekers. In 2024, Lake County is appointed to be the 10th top county in the United States based on the county's performance, capital investment, and job creation.

The population of Lake County in 2023 was 708,760, up from 374,447 in 1969. It is home to a productive population in the 15-19 and 20-24 age groups, with a balanced gender distribution: 50.4 percent males and 49.6 percent females. The most significant population growth occurred between 1985 and 2005, with a 46.9 percent increase. After that, population growth began to slow due to rapid suburban expansion. The forecasting analysis showed that the population is projected to increase by approximately 6,190 residents each year, which means that in 2040, the population would be 814,700.

Lake County has high living standards, with residents having high levels of education, incomes of more than $150,000 per year, and low poverty levels compared to the state of Illinois. Moreover, the neighborhood is diverse, comprising at least five racial groups and distributed across the county.

Lake County serves as a job generator both within the region and relative to national trends. Using location quotient (LQ) analysis, it shows that its primary industry is military, but manufacturing generates income for the county's economy, and retail supports local employment. Using shift-share analysis, the transportation and warehousing sector is growing rapidly in Lake County, not only because of the growth effect and industry trend, but

also due to strong local competitiveness. Lake County exhibits a monocentric urban structure, with employment concentrated in the central business district.

Houses in Lake County remain affordable on average. However, affordability has declined since 2019, and the burden has fallen on renters, who face growing financial stress, while homeowners are less affected. In terms of accessibility, there is spatial inequality in job access. Transportation accessibility in Lake County is highest in the central business district (CBD) and declines with distance from employment centers.

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APPENDICES

TABLE 3 POPULATION PROJECTION USING FOUR METHODS

Year	Population	lnpop	Linear	Geometric	Constant-change	Percent-change
1969	374,447	12.833	n/a	n/a	n/a	n/a
1970	382,881	12.855	n/a	n/a	n/a	n/a
1971	385,152	12.861	n/a	n/a	n/a	n/a
1972	391,375	12.877	n/a	n/a	n/a	n/a
1973	391,621	12.878	n/a	n/a	n/a	n/a
1974	398,883	12.896	n/a	n/a	n/a	n/a
1975	408,007	12.919	n/a	n/a	n/a	n/a
1976	416,636	12.940	n/a	n/a	n/a	n/a
1977	426,525	12.963	n/a	n/a	n/a	n/a
1978	433,994	12.981	n/a	n/a	n/a	n/a
1979	440,720	12.996	n/a	n/a	n/a	n/a
1980	442,432	13.000	n/a	n/a	n/a	n/a
1981	449,826	13.017	n/a	n/a	n/a	n/a
1982	451,470	13.020	n/a	n/a	n/a	n/a
1983	455,347	13.029	n/a	n/a	n/a	n/a
1984	457,407	13.033	n/a	n/a	n/a	n/a
1985	465,821	13.052	n/a	n/a	n/a	n/a
1986	473,870	13.069	n/a	n/a	n/a	n/a
1987	482,608	13.087	n/a	n/a	n/a	n/a
1988	492,692	13.108	n/a	n/a	n/a	n/a
1989	505,632	13.134	n/a	n/a	n/a	n/a
1990	520,187	13.162	n/a	n/a	n/a	n/a
1991	532,119	13.185	n/a	n/a	n/a	n/a
1992	543,244	13.205	n/a	n/a	n/a	n/a
1993	554,697	13.226	n/a	n/a	n/a	n/a
1994	567,357	13.249	n/a	n/a	n/a	n/a
1995	580,911	13.272	n/a	n/a	n/a	n/a
1996	596,182	13.298	n/a	n/a	n/a	n/a
1997	609,714	13.321	n/a	n/a	n/a	n/a
1998	621,618	13.340	n/a	n/a	n/a	n/a
1999	633,094	13.358	n/a	n/a	n/a	n/a
2000	648,150	13.382	n/a	n/a	n/a	n/a
2001	658,986	13.398	n/a	n/a	n/a	n/a
2002	668,022	13.412	n/a	n/a	n/a	n/a
2003	668,760	13.413	n/a	n/a	n/a	n/a
2004	677,735	13.427	n/a	n/a	n/a	n/a
2005	684,419	13.436	n/a	n/a	n/a	n/a

Year	Population	lnpop	Linear	Geometric	Constant-change	Percent-change
2006	689,891	13.444	n/a	n/a	n/a	n/a
2007	692,995	13.449	n/a	n/a	n/a	n/a
2008	697,369	13.455	n/a	n/a	n/a	n/a
2009	701,042	13.460	n/a	n/a	n/a	n/a
2010	704,663	13.465	n/a	n/a	n/a	n/a
2011	704,078	13.465	n/a	n/a	n/a	n/a
2012	706,490	13.468	n/a	n/a	n/a	n/a
2013	710,574	13.474	n/a	n/a	n/a	n/a
2014	712,615	13.477	n/a	n/a	n/a	n/a
2015	714,886	13.480	n/a	n/a	n/a	n/a
2016	716,547	13.482	n/a	n/a	n/a	n/a
2017	716,912	13.483	n/a	n/a	n/a	n/a
2018	716,484	13.482	n/a	n/a	n/a	n/a
2019	715,105	13.480	n/a	n/a	n/a	n/a
2020	713,281	13.478	n/a	n/a	n/a	n/a
2021	712,570	13.477	n/a	n/a	n/a	n/a
2022	709,899	13.473	n/a	n/a	n/a	n/a
2023	708,760	13.471	n/a	n/a	708,760	708,760
2024	n/a	n/a	786,329	824,252	714,951	717,184
2025	n/a	n/a	793,869	835,637	721,142	725,709
2026	n/a	n/a	801,408	847,180	727,333	734,335
2027	n/a	n/a	808,948	858,882	733,524	743,063
2028	n/a	n/a	816,488	870,745	739,715	751,895
2029	n/a	n/a	824,028	882,773	745,906	760,833
2030	n/a	n/a	831,568	894,966	752,097	769,876
2031	n/a	n/a	839,108	907,328	758,288	779,027
2032	n/a	n/a	846,648	919,861	764,479	788,286
2033	n/a	n/a	854,187	932,567	770,670	797,656
2034	n/a	n/a	861,727	945,448	776,861	807,137
2035	n/a	n/a	869,267	958,508	783,052	816,731
2036	n/a	n/a	876,807	971,747	789,243	826,439
2037	n/a	n/a	884,347	985,170	795,434	836,262
2038	n/a	n/a	891,887	998,778	801,625	846,202
2039	n/a	n/a	899,427	1,012,574	807,816	856,260
2040	n/a	n/a	906,966	1,026,560	814,007	866,437

TABLE 4 EMPLOYMENT CENTER IN LAKE COUNTY

Lists	Tracts	Total Value of Employment Center	Employment Center Share (%)	Center Type
Cand6	17097864521* 17097864524	39,290	32.84	CBD

Lists	Tracts	Total Value of Employment Center	Employment Center Share (%)	Center Type
	17097864802			
Cand5	17097864510 17097864513 17097863902 17097864002*	27,586	23.06	Subcenter 1
Cand3	17097863702*	17,446	14.58	Subcenter 2
Cand2	17097861504 17097862603* 17097862604	12,122	10.13	Subcenter 3
Cand1	17097864408* 17097864409	10,110	8.45	Subcenter 4
Cand8	17097865400*	6,778	5.67	Subcenter 5
Cand7	17097864904*	3,485	2.91	Subcenter 6
Cand4	17097863903*	2,828	2.36	Subcenter 7

* A tract with the highest percentile value within each candidate group

Density gradient function estimation results

	Dependent variable:			
	log(eden)		log(pden)	
	(1)	(2)	(3)	(4)
cbd_dist	-0.110***	-0.075***	-0.016	0.024
	(0.016)	(0.021)	(0.012)	(0.015)
subc_dist		-0.074**		-0.084***
		(0.030)		(0.022)
Constant	0.756***	0.722***	1.395***	1.358***
	(0.191)	(0.188)	(0.142)	(0.136)
Observations	158	158	158	158
R^2	0.244	0.272	0.012	0.098
Adjusted R^2	0.239	0.263	0.005	0.086
Residual Std. Error	1.062 (df = 156)	1.045 (df = 155)	0.789 (df = 156)	0.757 (df = 155)
F Statistic	50.236*** (df = 1; 156)	28.960*** (df = 2; 155)	1.841 (df = 1; 156)	8.377*** (df = 2; 155)
Note:				*p<0.1; **p<0.05; ***p<0.01

FIGURE 32 DENSITY GRADIENT FUNCTION ESTIMATION RESULTS

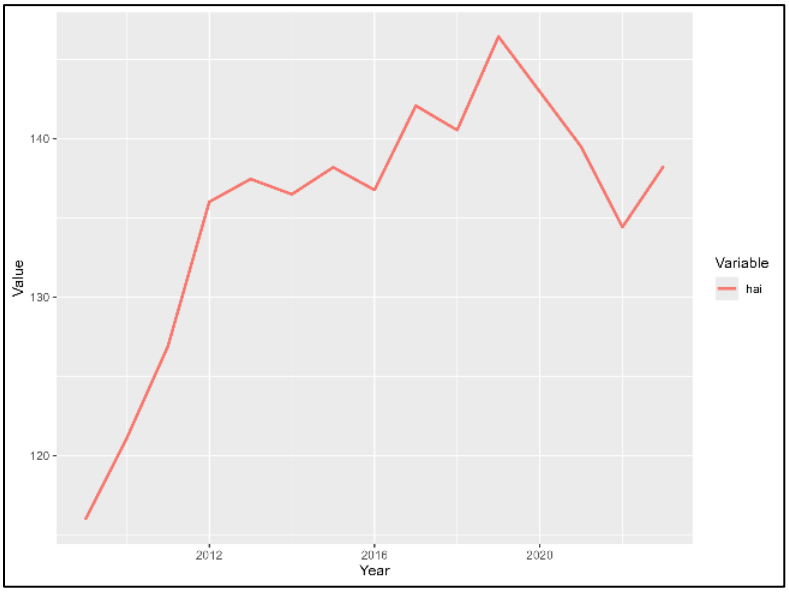

FIGURE 33 HOUSING AFFORDABILITY INDEX GROWTH IN 2009-2023

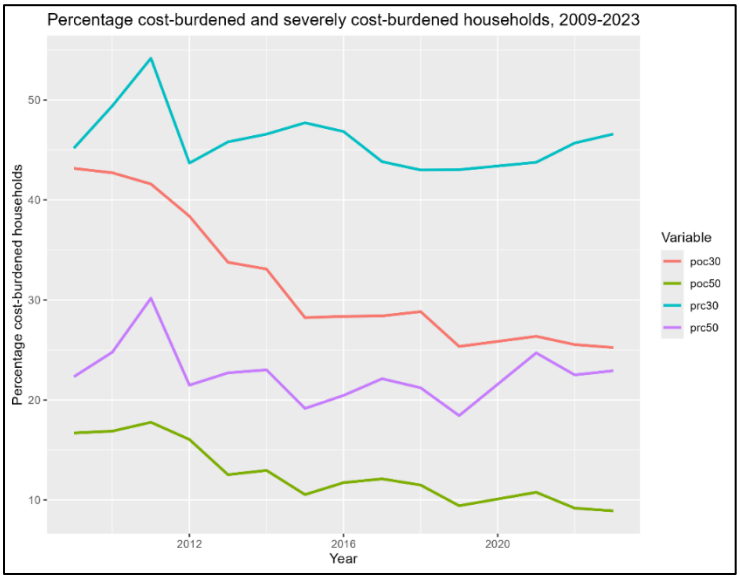

FIGURE 34 MEDIAN HOUSEHOLD INCOME AND HOME VALUE IN LAKE COUNTY FROM 2009-2023